Project Design Document

10/18/26
Christian

Project Concept

1 Player Control

You control a *Mini Sized Person* in this *3rd person 3D Wave Survival* game

where *WASD Arrow Keys* makes the player *Move*

2 Basic Gameplay

During the game, *Enemies(Insects/ Bugs)* appear from *All angles*

and the goal of the game is to *Defeat enemies, earn points and survive as long as possible.*

3 Sound & Effects

There will be sound effects *Attacks, damage and enemy defeat.* and particle effects *Enemy destruction*

[*optional*] There will also be *UI SFX/ VFX*

4 Gameplay Mechanics

As the game progresses, *More enemies/ types of enemies spawn,* making it *More difficult for the player to survive*

[*optional*] There will also be *Player abilities to help player evade, reposition and set up attacks.*

5 User Interface

The *Player Health* will *decrease* whenever *Player takes damage*

At the start of the game, the title *THEY BE BUGGIN* will appear and the game will end when *Player health reaches 0*

6 Other Features

Sound/ Visual effects/ Gameplay polish/ Assets/ Optimization/ Overall Polish can be added later.

Project Timeline

Milestone	Description	Due
#1	- *Add primitives(Floor, Walls, Player, Enemies), Camera and player movement*	*TODAY*
#2	- *Create enemy parent class and child classes ([EnemyParent]-Ant-Beetle-Wasp)*	*TODAY*
#3	- *Implement player attacks, enemy damage, health, and enemy defeat.*	*TODAY*
#4	*Implement enemy spawning, wave progression, and score tracking.*	*N/A*
#5	*Add title screen/ Game Over screen and UI; test the game, commit and merge a GitHub branch, and prepare the submission.*	*N/A*
Backlog	*Sound effects, particle effects, insect models, animations, and arena obstacles may be added later.*	*N/A*

Project Sketch

Swarm Survival

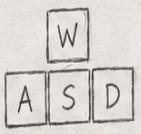
Game Concept

- Survive waves of bugs.
- Fight in a closed arena.
- Get points for defeating enemies.
- Survive as long as possible.

Gameplay Loop

1. Player moves around arena.
2. Enemies spawn in waves and chase player.
3. Player attacks and defeats enemies.
4. Earn points.
5. Next wave spawns (more enemies / harder).
6. Game ends when player health reaches zero.

Controls (PC)



Move

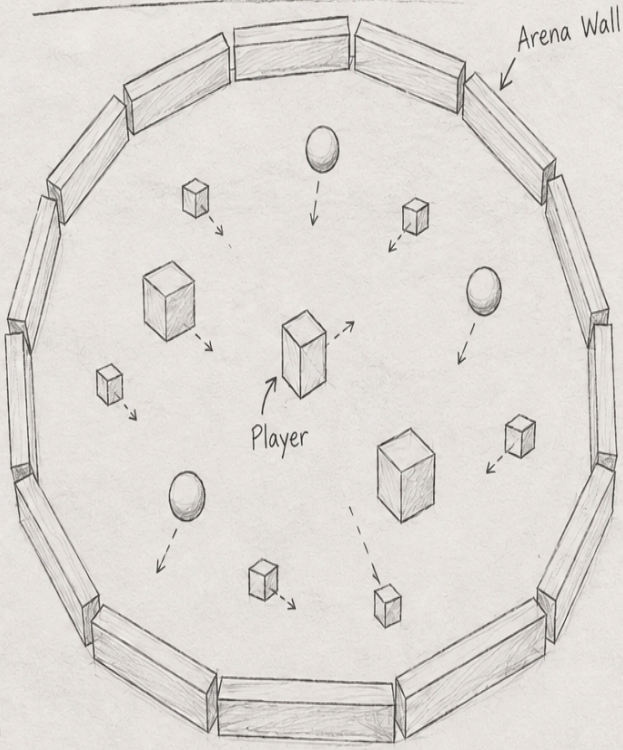


Left Click
Attack

Do we need obstacles?

- Not required.
- A simple open arena works.
- Easier to develop and focus on core gameplay and OOP.
- Can add obstacles later if desired for more challenge and variety.

Arena Sketch (Top-Down View)



Player Sketch (Primitive Objects)



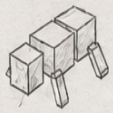
Front View



Side View

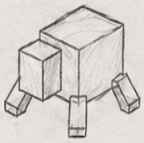
Enemy Sketches (Primitive Objects Only)

Ant (Fast)



- Small (cubes)
- Fast movement
- Low health
- Melee attack
- Spawns in large numbers

Beetle (Tank)



- Large (cubes)
- Slow movement
- High health
- High damage
- Spawns in smaller numbers

Wasp (Flying)



- Sphere body
- Flies (ignores ground)
- Medium health
- Ranged attack (shoots)
- Spawns in moderate numbers

UI Sketch (Screen View)

